PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004
MASTER OF COMPUTER APPLICATIONS SEMESTER : 1
23MX13 DATA STRUCTURES
TUTORIAL-2

MARKS: 25 Marks

1) Write an algorithm to perform the following:
   k) Count the number of nodes in a singly linked list
   l) Concatenate the shorter list at the end of a longer list
   m) Reverse the singly linked list
   n) Delete the node at the end of the list
   o) Delete the node at the specified position.
2) Write algorithms to i) Push and ii) Pop an element from a stack using an array
3) Write an algorithm to reverse a string using the stack.

PSG COLLEGE OF TECHNOLOGY
DEPARTMENT OF COMPUTER APPLICATIONS
20MX13 - DATA STRUCTURES
MCA – SEMESTER – I
TUTORIAL- 1

Date: 16-11-2022 **Marks: 25 Marks**

1) What are the characteristic properties of an algorithm? Check whether the below algorithm follows all the properties.
   Step 1: Start
   Step 2: Declare variables n factorial and i.
   Step 3: Initialize variables
   factorial←1
   i←1
   Step 4: Read value of n
   Step 5: Repeat the steps until i=n
   5.1: factorial←factorial*i
   Step 6: Display factorial
   Step 7: Stop
2) Write an algorithm for linear search. Describe the best and worst case complexities.
3) Write a recursive algorithm to perform binary search. Indicate best case and worst case complexity.
4) Consider a band matrix as shown in fig. 1, where elements on, above and below the main diagonal are non-zero and other elements are zero. Suggest an efficient storage representation and write the corresponding addressing function.

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 3 & 4 & 5 & 0 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & 9 & 11 \end{bmatrix}$$

$\frac{2(3)}{2}$

Fig.1

5) Write an algorithm to convert an infix expression into its equivalent postfix expression using stack. Trace the algorithm with the expression A + (B * C / D) ^ E.

PSG COLLEGE OF TECHNOLOGY
DEPARTMENT OF COMPUTER APPLICATIONS
20MX13- DATA STRUCTURES
MCA – SEMESTER – I
TUTORIAL- 2

Date:25-11-2022 **Marks: 25 Marks**

1) What is abstraction? What is an ADT? Why do we use ADT? Define String ADT
2) Suggest an efficient storage representation for storing a lower triangular matrix. Write the corresponding addressing function.
3) Write algorithms to i) Push and ii) Pop an element from a stack implemented using an array.
4) Formulate an algorithm to evaluate the postfix expression. Trace the algorithm on the expression a b + c d - * e f - / where a=2, b=2, c=4, d=1, e=3 and f=1
5) What is a priority queue? Give an application that demands the use of a priority queue.

ab+cd-*ef-/

e.g. 4+4-* 3-/

a=2
b=2
c=4
d=1
e=3
f=1

No of Pages : 2 Course Code : 23MX13

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004
MASTER OF COMPUTER APPLICATIONS Semester : 1
23MX13 DATA STRUCTURES

Time : 1 Hour 30 mins **Maximum Marks : 50**

INSTRUCTIONS:				
1. Answer **ALL** questions. Each question carries 25 Marks.				
2. Course Outcome Table:	Qn.1	CO1	Qn.2	CO2

1) a. i) What are the characteristic properties of an algorithm? **(2)**

ii) Write an algorithm to find the quotient of m/n given m and n. Do not use division operator. What is the complexity of your algorithm? **(3)**

b. i) Differentiate linear and non-linear data structures. Give examples. **(3)**

ii) Define Big O. How is it related to algorithm computational complexity? **(5)**

If $f(n) = O(g_1(n))$ and $h(n) = O(g_2(n))$, answer the following questions. Justify your answer

a) What is $O(f(n).h(n))$?

b) What is $O(f(n)+h(n))$?

c) If $g_1(n)) = g_2(n)$, Is $f(n)=h(n)$?

c. i) Indicate the complexity of the following code segments using Big O notation. **(6)**

```
1) for ( j = n/2; j ≤ n; j++)
       for( k =1; k ≤ n; k = 2 * k)
           for (m =1; m ≤ n ; m = m *2)
                   printf("#")
```

```
2) for(i=0; i<=n; i=i++)
       for (j= i; j <= i*i; j=j++)
           if(j%2 ==0) {
            for (k=0; k< j; k=k ++)
                 count ++;}
```

```
3) int fun(int n){
       int count = 0;
       for (int i = n; i > 0; i /= 2)
           for (int j = 0; j < i; j++)
               count += 1;
       return count;  }
```

```
4) for( int i = 0; i <n; i ++ ) {
       for( int j = 0; j < n; j *= 2 )
           printf("*");
       break;
   }
```

ii) Solve the following recurrence relation, using substitution method **(6)**

$$T(n)=2T(n/2)+n;$$
$$T(1)=1.$$

Using Mater Theorem, verify the solution.

2) a. Describe static and dynamic memory allocation. What are their advantages and limitations? **(5)**

b. i) Write a code snippet to count the number of nodes in a singly linked list. **(3)**

$n^k \log^p b$ $T(n) = aT(\frac{n}{b}) + n^k \log^p n$

$a = 2$ $b = 2$ $k = 1$ $p = 0$

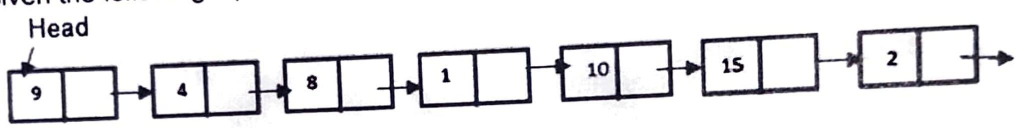
$(n^k \log^{p+1} n)$ $b^k = 2$

$O(n^k \log \log n)$

ii) What is the functionality of the following code snippet?

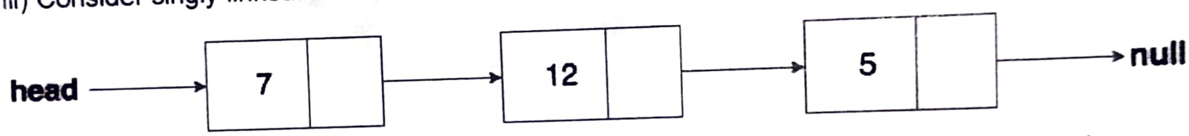
```
count = 0;
while (head != NULL)
  {
     if (count % 2 == 0)
        print(data);
     count++;
     head = head->next;
  }
```

Given the following input list, predict the output.

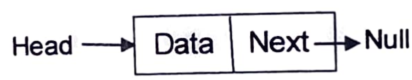


(3)

iii) Consider singly linked list in the format shown below:



Fill up the blank lines in the following code segment to print the sum of elements in the given linked list.




```
struct Node {
   int data;
   struct Node* next;
};
   int sum = 0;
   while (-----------) {
      ------------------
      ------------------  }
   print (sum) ;
}
```

c. i) How are two dimensional arrays represented in memory? Write the corresponding addressing functions. Extend this to represent the k-dimensional array. **(4)**

ii) Suggest an efficient way of storing the tri-band matrix and write the corresponding addressing function. A sample band matrix of order 4 x 4 is shown below: **(4)**

$$\begin{matrix} 10 & 5 & 0 & 0 \\ 9 & 6 & 5 & 0 \\ 0 & 1 & 7 & 3 \\ 0 & 0 & 19 & 2 \end{matrix}$$

iii) A Toeplitz matrix is a diagonal constant matrix i.e. all elements along a diagonal have the same value. Following is an example of Symmetric Toeplitz matrix. Suggest an efficient representation for storing Symmetric Toeplitz matrix. Write the corresponding addressing function. **(4)**

$$\begin{matrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 2 & 3 \\ 4 & 3 & 2 & 1 & 2 \\ 5 & 4 & 3 & 2 & 1 \end{matrix}$$

./END/

No of Pages : 2

Course Code : 23MX13

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004
MASTER OF COMPUTER APPLICATIONS **Semester : 1**
23MX13 **DATA STRUCTURES**
Test-2

Time : 1 Hour 30 mins
Maximum Marks : 50
Date: 21-12-2023

INSTRUCTIONS:
1. Answer **ALL** questions. Each question carries 25 Marks.
2. Course Outcome Table:

Qn.1	CO3	Qn.2	CO4

1) **a.** i) Construct a binary tree whose inorder traversal sequence is G D H B E I A F J C and the preorder traversal sequence is A B D G H E I C F J **(2)**

ii) Construct a strictly binary tree where the non-leaf nodes have 2 children and the preorder traversal sequence as 30, 20, 15, 5, 18, 25, 40, 35, 50, 45, 60 and postorder traversal sequence is 5, 18, 15, 25, 20, 35, 45, 60, 50, 40, 30 **(3)**

b. i) Identify suitable data structures for the following: **(2)**
1. Priority Queue
2. Representing contacts in social media
3. Representing mathematical expressions
4. Representing the file directories.

ii) Construct a binary search tree of level 3 with the following sequence of numbers: 10, 12, 5, 4, 20, 8, 7, 15 and 13. **(3)**

iii) What are the different ways of representing a Graph in memory? Illustrate with the graph shown in Fig. 1. **(3)**

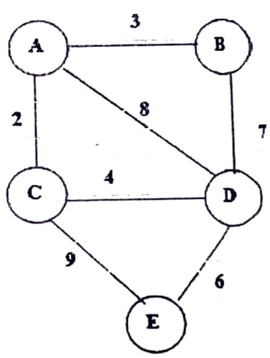


Fig.1

c.i) Write an algorithm for traversing a Graph using Depth First Search method. Trace the algorithm with the graph given in Fig. 2. **(6)**

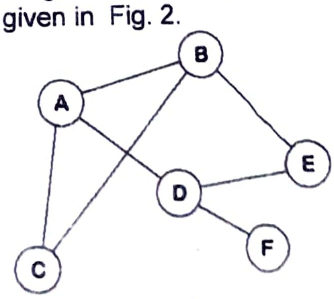


Fig.2

c.ii) Write an algorithm to perform deletion in Binary Search Tree. Consider the tree given in fig. 3, show the tree after deleting 8, 12, 6 and 10.

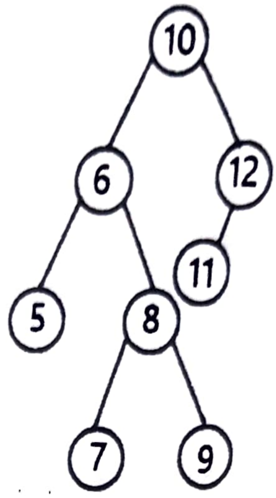


Fig.3

2) a. i) Illustrate Heap sort with the following elements: 34, 67, 89, 12, 45, 65, 90. What is the complexity?

b.i) Formulate linear search algorithm. Trace the algorithm on a sample input. Indicate the best and worst case complexity. (4

b. ii) Write an algorithm for Insertion Sort. Applying the algorithm to sort the following list of elements 12, 25, 3, 45, 56, 34, 70, 27, 189, 59, 75. Show the results of the intermediate passes, Analyze the best and worst case complexity. (4)

c. What is Hashing? What is the advantage and limitation of using Hashing to search an element? Explain the methods of resolving collisions. Consider a hash table of size 11 with hash function h(k)=k mod 11. Draw the table that results after inserting in the given order, the following values: 9, 26, 50, 15, 2, 21, 36, 22 and 31. Resolve collisions using each of the collision resolving techniques discussed. Describe insertion, searching and deletion in each of the methods discussed (12)

(OR)

c. For a binary tree, write an algorithm for the following: (12)

i) To count number of leaf nodes.

ii) To find the level of a node with the given value

iii) To check whether two trees are structurally identical.

/END/

5 15 18 20 25 30 35 40 45 50 60

30 20 15 5 18 25 40 35 50 45 60

No of Pages : 2 Course Code : 20MX13

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004
MASTER OF COMPUTER APPLICATIONS Semester : 1
20MX13 DATA STRUCTURES
TEST-2

Time : 1 Hour 30 mins **Maximum Marks : 50**

INSTRUCTIONS:				
1. Answer **ALL** questions. Each question carries 25 Marks.				
2. Course Outcome Table :	Qn.1	CO3	Qn.2	CO4

1) **a.** Write an algorithm i) to reverse a singly linked list and ii) to count number of nodes in a circular linked list. (5)

b. Given two polynomials represented by linked list, formulate algorithm for addition of two such polynomials. Trace the algorithm on a suitable example. (8)

c. Write an algorithm i) to merge two singly linked list and ii) for insertion and deletion of a node to the right of node pointed by M in doubly linked list. (12)

2) **a.** Define the following and give an example for each. (5)

i) ~~complete binary tree~~

ii) strictly binary tree

iii) skew tree.

b. **i)** Represent the binary tree shown in fig. 1 using i) array ii) linked list. Write the inorder, preorder and postorder traversal sequence. (4)

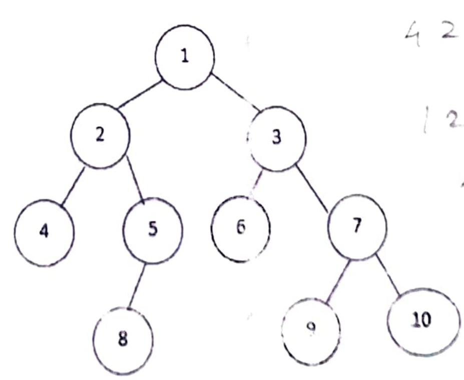


Fig. 1

ii) Draw the binary tree whose inorder traversal sequence is CBHDAFGE and preorder traversal sequence is ABCDHEFG (4)

c. Consider keys 45, 23, 11, 15, 25, 36, 20 and a hash table of size 11. Use hash function h(K)= K mod 11 and construct hashing tables with i) linear probing ii) quadratic probing and iii) double hashing for collision resolution. Illustrate how search is performed in each case.

(12)

(OR)

c. Write algorithms for BFS and DFS. Trace the algorithms on the graph shown in fig.2

(12)

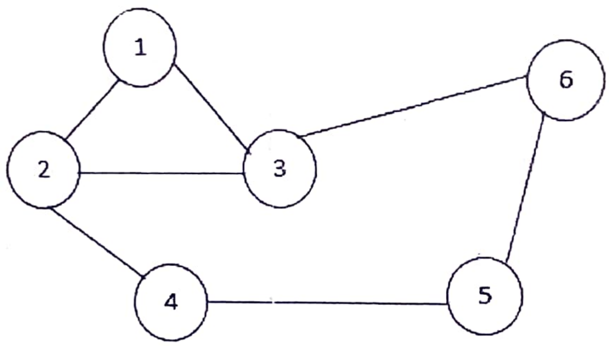


Fig. 2

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No of Pages : 2

Course Code : 20MX13

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004
MASTER OF COMPUTER APPLICATIONS   Semester : 1
20MX13   DATA STRUCTURES

Time : 1 Hour 30 mins

Maximum Marks : 50

INSTRUCTIONS:				
1. Answer **ALL** questions. Each question carries 25 Marks.				
2. Course Outcome Table	Qn.1	CO1	Qn.2	CO2

1) **a.** What are the characteristic properties of an algorithm? Write an algorithm to find Ka without using multiplication, where K and a are integers. 2 *3 6 (5)

b. Explain various ways of representing two-dimensional arrays in memory. Give suitable examples. Consider a tri diagonal matrix as shown in fig. 1, where elements on, above and below the main diagonal are non-zero and other elements are zero. Suggest an efficient storage representation and write the corresponding addressing function. (8)

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 3 & 4 & 5 & 0 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & 9 & 11 \end{bmatrix}$$

(2i+j)   4 +4 - * 3 - /

Fig.1

8/

c. Write algorithms for **i)** Linear and **ii)** Binary Search. Trace the algorithm on a sample input. Indicate the best and worst case complexity of the algorithms. (12)   2/

2) **a.** Write algorithms for insert and delete operations on Circular Queue. (5)

b. Formulate an algorithm to evaluate the postfix expression. Trace the algorithm on the expression a b + c d - * e f - / where a=2, b=2, c=4, d=1, e=3 and f=1 (8)

2 2 4 1 3 1

c. Write an algorithm to check the well- formedness of parentheses in an expression using stack.Use the primitive operations of stack and trace the algorithm on the inputs

i) (a/((b+c)*c-d)/e)   abcd

ii) (a+b)*(e-b   ab (12)

(OR)

c. Write an algorithm to convert an infix expression into its equivalent postfix expression using stack. Trace the algorithm with the expression: A+ (B* C / D) ^ E. (12)

aB*C/D+^E

/END/

No of Pages : 3

4562

Course Code : 23MX13

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, DECEMBER 2023

MCA Semester : 1

23MX13 DATA STRUCTURES

Time : 3 Hours **Maximum Marks : 100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Subdivision (a) carries 5 marks each, subdivision (b) carries 8 marks each and subdivision (c) carries 12 marks each.
3. If any data is found missing, suitable assumptions can be made with proper justification.
4. Course Outcome : Table

Qn.1	CO1	Qn.2	CO2	Qn.3	CO3	Qn.4	CO4

1. a) i) What is an ADT? What is its need? Define Queue ADT. (5)

   b) What are the characteristics properties of an algorithm? Write an algorithm for printing the first n elements in the Fibonacci sequence. Show that your algorithm possess all the properties of an algorithm. (8)

   c) i) Indicate the complexity of the following code segments using Big O notation. (4)

```
1) for ( j = 1; j ≤ n; j++)
       for( k = n; k > 1; k = k/2)
           for (m =1; m ≤ n ; m = m *2)
               printf("$")
```

```
2) for(j = 1; j < n; j++)
       for( k = 1; k < n ; k++) {
           print("$")
           break;
       }
```

   ii) Define Big O notation. Write the significance of Big O notation. Solve the following recurrence relation, using substitution method and write the complexity using Big O notation (8)

$$T(n) = T(n-1) + T(n-2) + c$$

$$T(1) = 1$$

2. a) What are the various ways of representing two dimensional matrix in storage? Explain with suitable examples. Write the corresponding addressing functions. Suggest an efficient representation for storing an upper triangular matrix and write the addressing function. (5)

   b) i) Write an algorithm using Stack ADT calls to recognize strings with balanced parentheses and square brackets. Trace the algorithm with strings ([( )]), (( )]) (4)

   ii) Implement a queue using a singly linked list. (4)

c) i) Consider the ADT Set that represents a collection of Integers. The ADT should support standard set operations such as the following:

i. S = init( ) /* Initialize S to the empty Set */

ii. isSingleton(S) /* Return True if and only if S is an empty Set */

iii. isMember(S,a) /* Return True if and only if a is a member of the Set S */

iv. S = addElement(S,a) /* Add a to the Set S. If a is already in S, there will no change in S */

v. S = delElement(S,a) /* Remove a from the Set S. If a is not a member of S, there will no change */

vi. Print(S) /* Print the elements of S */

Implement the Set ADT using Linked Lists (12)

(OR)

ii) Formulate algorithms for the following

1) Given two sorted list, merge them as a singly linked list

2) Given a doubly linked list and a key, print the data stored in the predecessor and successor nodes of the key and delete the key element. If the key element is not present, print an appropriate message.

3. a) Write the sequence in which nodes are traversed in inorder, preorder and postorder traversal on Binary Trees. Draw a binary tree whose inorder traversal sequence is H D J B E A F J C G and post order traversal sequence is H J D E B J F G C A. (5)

Draw a Binary Tree in which all the non-leaf nodes have two children and whose pre order traversal sequence is 100, 34, 16, 9, 8, 38, 11, 4, 81 and post order traversal sequence is 34, 9, 11, 4, 38, 81, 8, 16, 100.

b) i) Define heap data structure. Why is heap a preferred data structure to implement priority queue? Illustrate the max heap construction with the keys 34, 56, 17, 12, 8, 3, 2, 1, 0. What is the time complexity for inserting an element into the heap? (5)

ii) Draw a) directed and b) undirected graph with 5 vertices and 6 edges. Show their representation using Adjacency matrix and Adjacency list (3)

c) i) Write algorithms for Breadth First Search and Depth First Search on Graphs. Tracing the algorithms, write the traversal sequence for the graph given in fig.1. Modify any one of the algorithms to check whether the given graph is connected. (12)

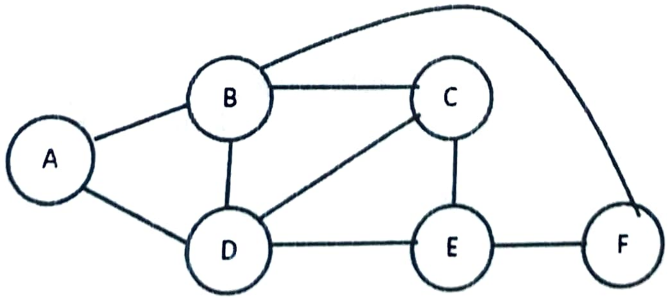

Fig.1

(OR)

ii) What is a Binary Search Tree? Formulate algorithms for insertion and deletion on a Binary Search Tree. Create a Binary Search tree of level 4 with the following elements : 45, 65, 78, 90, 23, 30, 10, 50, 5, 10, 60, 12

4. a) Illustrate heap sort and radix sort with the following numbers. 45, 65, 78, 90, 23, 30, 10, 50. Show the result of the intermediate passes. What is the complexity of these algorithms? (5)

b) i) Formulate an algorithm for binary search. When Binary Search algorithm is applied to search for keys 67 and 15 in the list of elements 2, 4, 12, 22, 37, 39, 45, 48, 52, 64, 67, 69, 80, in which order elements will be compared with the keys. What is the time complexity for successful and unsuccessful search? (4)

ii) Formulate algorithm for Bubble sort. What is the complexity of Bubble sort algorithm. (4)

c) What is Hashing? Describe any two methods for finding the hash address. What is meant by Collision in hashing? Describe any two ways of resolving collisions. The following elements are to be entered in the given order in a hash table of size 11: 12, 31, 63, 89, 90, 78, 55 and 21. Apply a suitable hash function. Show the contents of the table. Illustrate collision resolution methods described. For each of the methods, how many comparisons are to be made in locating the elements: 31, 89 and 56. (12)

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No of Pages : 3

Course Code : 20MX13

Roll No:
(To be filled in by the candidate)

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PSG COLLEGE OF TECHNOLOGY, COIMBATORE 641 004

SEMESTER EXAMINATIONS, JANUARY 2023

MCA Semester : 1

20MX13 DATA STRUCTURES

Time : 3 Hours **Maximum Marks : 100**

INSTRUCTIONS:								
1. Answer **ALL** questions. Each question carries 25 Marks.								
2. Course Outcome Table :	Qn.1	CO 1	Qn.2	CO 2	Qn.3	CO 3	Qn.4	CO 4

1. a) What is Abstract data type (ADT)? Define String ADT. (5)

   b) How are multidimensional matrices stored in memory? Write the addressing function corresponding to each representation. A Toeplitz Matrix is a square matrix in which the elements on a diagonal are same. Following is an example of Toeplitz Matrix:

10	2	3	4	5
6	10	2	3	4
7	6	10	2	3
8	7	6	10	2
9	8	7	6	10

   Suggest an efficient storage representation for storing Toeplitz Matrix and write the corresponding addressing function. (8)

   c) Formulate algorithms for linear and binary search. Trace the algorithms on sample input to show the correctness of the algorithm. Write the best and worst case complexity of both algorithms. (12) 4

2. a) What is a circular Queue? Highlight the difference between linear and circular queue. Write algorithms for the primitive operations on circular queue. (5) 3

   b) Write an algorithm to check the well formedness of parenthesis in an expression. Consider (,), [,], { and } types of parentheses. Trace the algorithm on the inputs {a + [((c-d) * e)] } and {a + [c-(a+d])}. (8) 6

   c) i) Write algorithms for the primitive operations on array based stack. Formulate an algorithm converting an infix expression to postfix form using stacks. Trace the algorithm on the expression a * (b + c *d) / e. a bcd * + * e/ (12) 10

   (OR)

   ii) a. What is a dequeue? What are the different types of dequeue? (4)

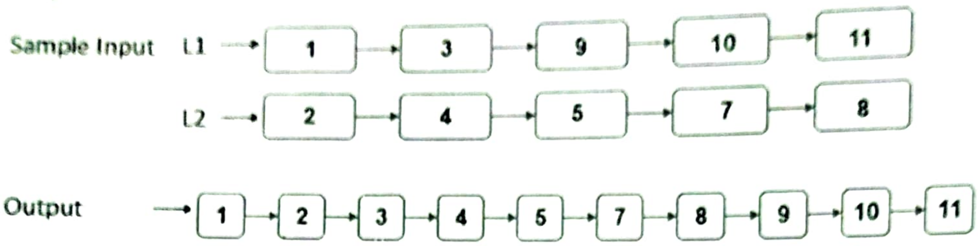
   b. Formulate an algorithm to evaluate postfix expressions. Trace the algorithm on the expression abc*+cd-* where a = 2, b = 4, c = 2, d = 1. (8)

3. a) i. Differentiate static and dynamic memory allocation. (2)

   ii. Write an algorithm to count the number of nodes in a circularly linked list. (3)

4 2 3 1 7

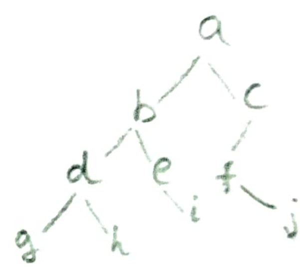
b) Formulate an algorithm to merge two sorted lists. Sample input and corresponding output are shown below: (8)



c) i) 1) How can a polynomial be represented using linked lists? Represent $5x^5 + 4x^3 + 3$ using linked lists. Write an algorithm to add two polynomials represented using linked lists. (8)

2) Represent the following sparse matrix as a linked list. (4)

0	0	3	0
0	0	0	8
1	0	3	0
0	0	7	0

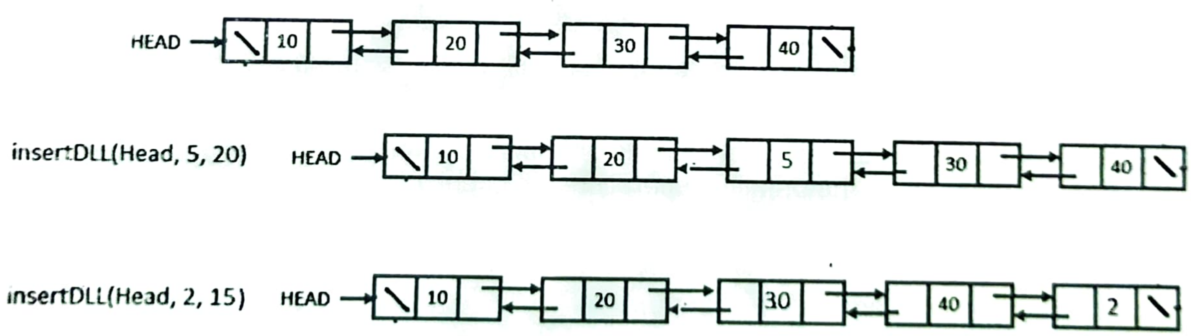


(OR)

ii) 1) What is a linked queue? Write algorithms for the primitive operations on linked queue. (6)

2) Write an algorithm insertDLL(Head, elt, value) to add elt, in a doubly linked list next to the given value if the value is present in the list. Otherwise add the element at the end of the list.

Sample inputs and outputs are shown below: (6)



4. a) Write the order in which the nodes are visited in the different types of traversals in a Binary tree. Draw the binary tree whose inorder traversal sequence is: g d h b e i a f j c and preorder traversal sequence is: a b d g h e i c f j. (5)

b) i) What are the different types of representation of graphs? Give an example for the following types of graphs and represent them using various types of representations :

Directed Graph, Undirected weighted graph. (4)

ii) Traverse the graph shown in Fig 1 using Breadth First Search (BFS) and Depth First Search (DFS) and write the traversal sequences. Indicate the data structure used in BFS and DFS. (4) 2

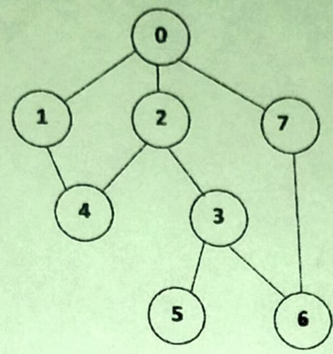


Fig 1

c) What is hashing? Give any two hash functions. What are collisions in hashing? What are the ways of resolving collisions? Describe how elements are located with different types of collision resolution methods. How can the hash performance be improved? Consider the following items to be inserted into a hash table : 69, 66, 80, 35, 18, 60, 89, 70, 12 and a hash table of size 11. Use any one hash function of your choice and illustrate all types of collision resolution techniques. (12) 7

/END/

FD/RL

dm

66
18
70
12
35
60
69
80
89

seq

18 - 12
35
60
69 -> 66 -> 70
80
89

6 -> 69
6 -> 66
7 -> 80
3 -> 35
1 -> 18
5 -> 60
8 -> 89
6 -> 70
1 -> 12